

Credit Card Fraud Detection

Abstract

This project addresses the growing concern of credit card fraud by developing a machine learning-based detection system. Utilizing a publicly available dataset of anonymized transactions, we preprocess, analyse, and model the data using multiple classification techniques. Through exploratory data analysis and strategic handling of class imbalance, we achieve a robust fraud detection pipeline. The results demonstrate that machine learning can play a vital role in identifying suspicious activities and enhancing financial security.

1. Introduction

Credit card fraud is one of the most pressing concerns in today's digital age. With the rise of online transactions, it becomes essential to develop systems that can accurately detect fraudulent activity and prevent potential financial losses. This project aims to build a machine learning model that can effectively distinguish between legitimate and fraudulent transactions using real-world credit card transaction data.

2. Dataset Overview

The dataset used in this project is sourced from a real-world anonymized credit card dataset. It contains transactions made by European cardholders in September 2013.

- **Total Records:** 284,807 transactions
- **Fraudulent Cases:** 492
- **Features:**
 - Time: Seconds elapsed between this transaction and the first transaction in the dataset.
 - V1 to V28: Principal Component Analysis (PCA) transformed features.
 - Amount: Transaction amount.
 - Class: Response variable (0 = normal, 1 = fraud).

3. Data Preprocessing

- Loaded data using Pandas: `pd.read_csv()`
- Displayed first few rows using `data.head()`
- Checked for missing values and data types
- Distribution of classes revealed high class imbalance (492 fraud cases out of 284,807)
- Applied scaling to 'Amount' feature where needed

4. Exploratory Data Analysis (EDA)

- Used visualizations (Matplotlib, Seaborn) to explore class distribution
- Plotted histograms of transaction amount and time
- Analyzed correlations using heatmaps
- Observed that frauds tend to occur at specific times and amounts

5. Model Building

Multiple classification models were trained and compared:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- XGBoost Classifier

To handle class imbalance:

- Undersampling the majority class
- Oversampling using SMOTE

6. Model Evaluation

Evaluation metrics used:

- **Confusion Matrix:** To visualize TP, TN, FP, FN
- **Precision & Recall:** Focused on high recall to minimize false negatives
- **F1 Score:** Balanced metric for imbalanced data
- **ROC AUC Curve:** Measured area under ROC for better threshold evaluation

Best performing model: **Random Forest** with high recall and F1 score after SMOTE balancing.

7. Conclusion

This project demonstrates the effectiveness of machine learning in detecting fraudulent transactions. Using PCA-transformed features and careful preprocessing, models such as Random Forest showed promising results in identifying fraud.

8. Future Improvements:

- Use deep learning models like Autoencoders for anomaly detection
- Incorporate real-time streaming data
- Explore ensemble techniques